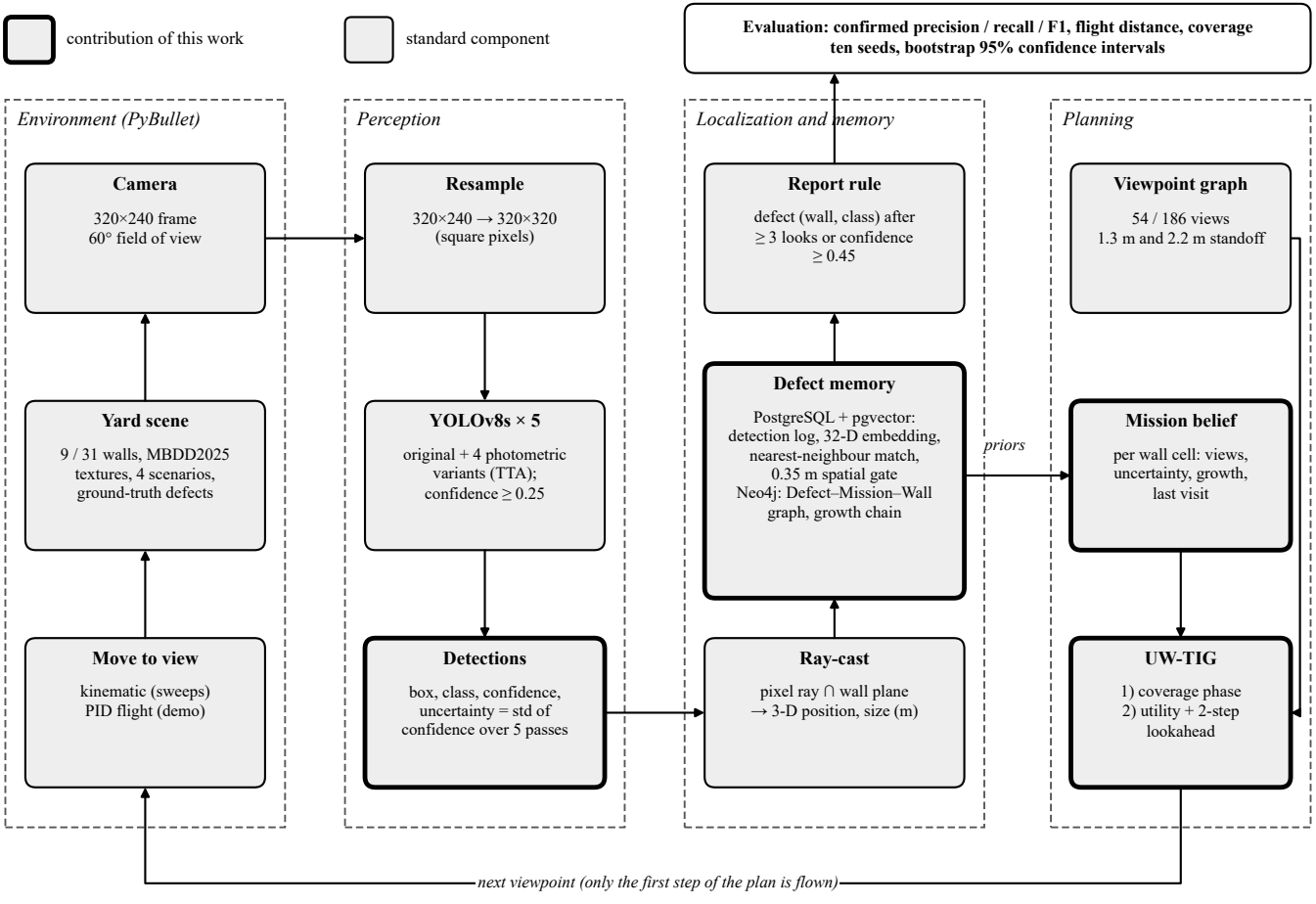

contribution of this work
standard component
Evaluation: confirmed precision / recall / F1, flight distance, coverage
ten seeds, bootstrap 95% confidence intervals
Environment (PyBullet)
Perception
Localization and memory
Planning
Camera
320×240 frame
60° field of view
Resample
320×240 → 320×320
(square pixels)
Report rule
defect (wall, class) after
≥ 3 looks or confidence
≥ 0.45
Viewpoint graph
54 / 186 views
1.3 m and 2.2 m standoff
Yard scene
9 / 31 walls, MBDD2025
textures, 4 scenarios,
ground-truth defects
YOLOv8s × 5
original + 4 photometric
variants (TTA);
confidence ≥ 0.25
Defect memory
PostgreSQL + pgvector:
detection log, 32-D embedding,
nearest-neighbour match,
0.35 m spatial gate
Neo4j: Defect–Mission–Wall
graph, growth chain
priors
Mission belief
per wall cell: views,
uncertainty, growth,
last visit
Move to view
kinematic (sweeps)
PID flight (demo)
Detections
box, class, confidence,
uncertainty = std of
confidence over 5 passes
Ray-cast
pixel ray ∩ wall plane
→ 3-D position, size (m)
UW-TIG
1) coverage phase
2) utility + 2-step
lookahead
next viewpoint (only the first step of the plan is flown)